

DAC Enode Intelligence Watcher - Custom All Time Report

Generated at UTC: 2026-09-12T14:08:27.380123+00:00

Report range: All Time

Project: DAC Enode Intelligence Watcher

This custom report is generated from local watcher JSON outputs.

Important note: this report is observation-based. It does not make official DAC ownership claims and should not be treated as a definitive decentralization measurement.

1. Report Scope

Field	Value
Range	All Time
Observation count	195
First observed source time	May 15, 2026 (00:00 CEST)
Last observed source time	2026-09-12 13:00 CEST
Latest watcher checked_at_utc	2026-09-12T11:06:04.471597+00:00
Latest source generated time	2026-09-12 13:00 CEST

2. Enode Movement Summary

Metric	Value
Minimum enode count	2
Maximum enode count	16
Average enode count	10.78
Total added observations	181
Total removed observations	160

Metric	Value
Target ports observed	28657

3. Observation Source Coverage

Phase	Observations
automated_watcher	181
manual_backfill	14

4. Anomaly Summary

Field	Value
Selected anomaly signals	20
Global anomaly summary	20 anomaly signals were detected across 195 observations. The highest observed anomaly severity is CRITICAL.
Global highest severity	CRITICAL
Recommended action	Use these anomaly events as candidates for deeper manual review and future technical reporting.

Severity	Signals in selected range
HIGH	12
CRITICAL	4
MEDIUM	4

Anomaly Type	Signals in selected range
HIGH_REMOVAL_EVENT	5
AGGRESSIVE_ROTATION	5
WATCHER_HIGH_SEVERITY_SIGNAL	3
SHARP_ENODE_COUNT_DROP	2
LOW_CONTINUITY_RATIO	2

Anomaly Type	Signals in selected range
MODERATE_ROTATION_SPIKE	2
LARGE_ENODE_COUNT_DROP	1

5. Provider / ASN Concentration Context

Field	Value
Overall concentration label	MODERATE
Headline	Observed infrastructure shows moderate concentration under the current heuristic model.
Key observation	Top live ASN is AS51167 with 16 unique IPs (36.36%).
Country observation	Top live ASN country code is DE with 19 unique IPs (43.18%).
Interpretation	Top live ASN controls at least 35% of observed unique IPs.
Disclaimer	Provider concentration and decentralization risk summary is an observation-based heuristic. It is based on currently available watcher data, live ASN enrichment, static provider hints, and DAC Infrastructure Signal labels. It should not be treated as an official DAC classification or as a definitive decentralization measurement.

Dimension	Top Name	Top Count	Top %	Label
Live ASN	AS51167	16	36.36	MODERATE
Live Country	DE	19	43.18	MODERATE
DAC Infrastructure Signal	Retained Infrastructure Signal	14	31.82	LOW

6. Observation Timeline

#	Phase	Status	Source Time	Current	Added	Removed	Unchanged	Port
1	manual_backfill	manual_initial	May 15, 2026 (00:00 CEST)	11	11	0	0	28657
2	manual_backfill	manual_changed	May 16, 2026 (20:00 CEST)	14	4	1	10	28657
3	manual_backfill	manual_changed	May 17, 2026 (16:00 CEST)	13	1	2	12	28657
4	manual_backfill	manual_changed	May 18, 2026 (08:00 CEST)	13	1	1	12	28657

#	Phase	Status	Source Time	Current	Added	Removed	Unchanged	Port
5	manual_backfill	manual_changed	May 19, 2026 (16:00 CEST)	7	1	7	6	28657
6	manual_backfill	manual_changed	May 20, 2026 (20:00 CEST)	12	5	0	7	28657
7	manual_backfill	manual_changed	May 22, 2026 (00:00 CEST)	12	1	1	11	28657
8	manual_backfill	manual_changed	May 23, 2026 (12:00 CEST)	12	4	4	8	28657
9	manual_backfill	manual_changed	May 24, 2026 (04:00 CEST)	9	0	3	9	28657
10	manual_backfill	manual_changed	May 25, 2026 (04:00 CEST)	12	3	0	9	28657
11	manual_backfill	manual_changed	May 28, 2026 (08:00 CEST)	14	5	3	9	28657
12	manual_backfill	manual_unchanged	May 29, 2026 (04:00 CEST)	14	0	0	14	28657
13	manual_backfill	manual_changed	May 30, 2026 (12:00 CEST)	14	1	1	13	28657
14	manual_backfill	manual_changed	May 31, 2026 (12:00 CEST)	15	2	1	13	28657
15	automated_watcher	initial	Jun 1, 2026 (04:00 CEST)	12	12	0	0	28657
16	automated_watcher	changed	Jun 1, 2026 (08:00 CEST)	13	1	0	12	28657
17	automated_watcher	changed	Jun 1, 2026 (12:00 CEST)	9	0	4	9	28657
18	automated_watcher	changed	Jun 1, 2026 (22:00 CEST)	10	1	0	9	28657
19	automated_watcher	changed	Jun 2, 2026 (06:00 CEST)	12	2	0	10	28657
20	automated_watcher	changed	Jun 2, 2026 (12:00 CEST)	12	1	1	11	28657
21	automated_watcher	changed	Jun 2, 2026 (22:00 CEST)	14	2	0	12	28657
22	automated_watcher	changed	Jun 3, 2026 (12:00 CEST)	10	0	4	10	28657
23	automated_watcher	changed	Jun 3, 2026 (14:00 CEST)	8	0	2	8	28657

#	Phase	Status	Source Time	Current	Added	Removed	Unchanged	Port
24	automated_watcher	changed	Jun 3, 2026 (22:00 CEST)	10	2	0	8	28657
25	automated_watcher	changed	Jun 4, 2026 (00:00 CEST)	11	1	0	10	28657
26	automated_watcher	changed	Jun 4, 2026 (02:00 CEST)	12	1	0	11	28657
27	automated_watcher	changed	Jun 4, 2026 (12:00 CEST)	11	0	1	11	28657
28	automated_watcher	changed	Jun 4, 2026 (18:00 CEST)	10	0	1	10	28657
29	automated_watcher	changed	Jun 6, 2026 (08:00 CEST)	11	1	0	10	28657
30	automated_watcher	changed	Jun 6, 2026 (10:00 CEST)	12	1	0	11	28657
31	automated_watcher	changed	Jun 6, 2026 (14:00 CEST)	13	1	0	12	28657
32	automated_watcher	changed	Jun 6, 2026 (16:00 CEST)	14	1	0	13	28657
33	automated_watcher	changed	Jun 6, 2026 (20:00 CEST)	15	1	0	14	28657
34	automated_watcher	changed	Jun 6, 2026 (22:00 CEST)	14	0	1	14	28657
35	automated_watcher	changed	Jun 7, 2026 (02:00 CEST)	15	1	0	14	28657
36	automated_watcher	changed	Jun 7, 2026 (08:00 CEST)	14	0	1	14	28657
37	automated_watcher	changed	Jun 7, 2026 (10:00 CEST)	15	1	0	14	28657
38	automated_watcher	changed	Jun 7, 2026 (14:00 CEST)	15	1	1	14	28657
39	automated_watcher	changed	Jun 7, 2026 (20:00 CEST)	16	1	0	15	28657
40	automated_watcher	changed	Jun 8, 2026 (06:00 CEST)	15	0	1	15	28657
41	automated_watcher	changed	Jun 8, 2026 (20:00 CEST)	14	0	1	14	28657
42	automated_watcher	changed	Jun 8, 2026 (22:00 CEST)	15	1	0	14	28657

#	Phase	Status	Source Time	Current	Added	Removed	Unchanged	Port
43	automated_watcher	changed	Jun 9, 2026 (00:00 CEST)	14	0	1	14	28657
44	automated_watcher	changed	Jun 9, 2026 (02:00 CEST)	12	0	2	12	28657
45	automated_watcher	changed	Jun 9, 2026 (06:00 CEST)	11	0	1	11	28657
46	automated_watcher	changed	Jun 9, 2026 (10:00 CEST)	10	0	1	10	28657
47	automated_watcher	changed	Jun 9, 2026 (12:00 CEST)	11	1	0	10	28657
48	automated_watcher	changed	Jun 9, 2026 (16:00 CEST)	10	0	1	10	28657
49	automated_watcher	changed	Jun 9, 2026 (20:00 CEST)	11	1	0	10	28657
50	automated_watcher	changed	Jun 9, 2026 (22:00 CEST)	12	1	0	11	28657
51	automated_watcher	changed	Jun 10, 2026 (06:00 CEST)	11	0	1	11	28657
52	automated_watcher	changed	Jun 10, 2026 (10:00 CEST)	12	1	0	11	28657
53	automated_watcher	changed	Jun 10, 2026 (14:00 CEST)	13	1	0	12	28657
54	automated_watcher	changed	Jun 10, 2026 (18:00 CEST)	14	1	0	13	28657
55	automated_watcher	changed	Jun 11, 2026 (00:00 CEST)	15	1	0	14	28657
56	automated_watcher	changed	Jun 11, 2026 (02:00 CEST)	14	0	1	14	28657
57	automated_watcher	changed	Jun 11, 2026 (08:00 CEST)	14	1	1	13	28657
58	automated_watcher	changed	Jun 11, 2026 (12:00 CEST)	13	0	1	13	28657
59	automated_watcher	changed	Jun 11, 2026 (20:00 CEST)	14	1	0	13	28657
60	automated_watcher	changed	Jun 11, 2026 (22:00 CEST)	15	1	0	14	28657
61	automated_watcher	changed	Jun 12, 2026 (16:00 CEST)	14	0	1	14	28657

#	Phase	Status	Source Time	Current	Added	Removed	Unchanged	Port
62	automated_watcher	changed	Jun 12, 2026 (20:00 CEST)	15	1	0	14	28657
63	automated_watcher	changed	Jun 12, 2026 (22:00 CEST)	16	1	0	15	28657
64	automated_watcher	changed	Jun 13, 2026 (00:00 CEST)	15	0	1	15	28657
65	automated_watcher	changed	Jun 13, 2026 (04:00 CEST)	16	1	0	15	28657
66	automated_watcher	changed	Jun 13, 2026 (08:00 CEST)	15	0	1	15	28657
67	automated_watcher	changed	Jun 13, 2026 (22:00 CEST)	14	0	1	14	28657
68	automated_watcher	changed	Jun 14, 2026 (02:00 CEST)	15	1	0	14	28657
69	automated_watcher	changed	Jun 14, 2026 (12:00 CEST)	14	0	1	14	28657
70	automated_watcher	changed	Jun 14, 2026 (14:00 CEST)	16	2	0	14	28657
71	automated_watcher	changed	Jun 14, 2026 (18:00 CEST)	14	0	2	14	28657
72	automated_watcher	changed	Jun 14, 2026 (20:00 CEST)	14	1	1	13	28657
73	automated_watcher	changed	Jun 15, 2026 (08:00 CEST)	13	0	1	13	28657
74	automated_watcher	changed	Jun 15, 2026 (20:00 CEST)	11	0	2	11	28657
75	automated_watcher	changed	Jun 16, 2026 (04:00 CEST)	10	0	1	10	28657
76	automated_watcher	changed	Jun 16, 2026 (14:00 CEST)	11	2	1	9	28657
77	automated_watcher	changed	Jun 16, 2026 (18:00 CEST)	8	0	3	8	28657
78	automated_watcher	changed	Jun 16, 2026 (22:00 CEST)	9	1	0	8	28657
79	automated_watcher	changed	Jun 17, 2026 (06:00 CEST)	10	1	0	9	28657
80	automated_watcher	changed	Jun 17, 2026 (12:00 CEST)	2	0	8	2	28657

#	Phase	Status	Source Time	Current	Added	Removed	Unchanged	Port
81	automated_watcher	changed	Jun 17, 2026 (16:00 CEST)	6	5	1	1	28657
82	automated_watcher	changed	Jun 17, 2026 (18:00 CEST)	7	1	0	6	28657
83	automated_watcher	changed	Jun 17, 2026 (20:00 CEST)	7	1	1	6	28657
84	automated_watcher	changed	Jun 18, 2026 (00:00 CEST)	8	1	0	7	28657
85	automated_watcher	changed	Jun 18, 2026 (08:00 CEST)	7	0	1	7	28657
86	automated_watcher	changed	Jun 18, 2026 (13:00 CEST)	10	3	0	7	28657
87	automated_watcher	changed	Jun 18, 2026 (16:00 CEST)	9	1	2	8	28657
88	automated_watcher	changed	Jun 18, 2026 (19:00 CEST)	8	0	1	8	28657
89	automated_watcher	changed	Jun 19, 2026 (00:00 CEST)	9	1	0	8	28657
90	automated_watcher	changed	Jun 19, 2026 (07:00 CEST)	8	0	1	8	28657
91	automated_watcher	changed	Jun 19, 2026 (11:00 CEST)	6	1	3	5	28657
92	automated_watcher	changed	Jun 19, 2026 (15:00 CEST)	9	3	0	6	28657
93	automated_watcher	changed	Jun 19, 2026 (18:00 CEST)	9	1	1	8	28657
94	automated_watcher	changed	Jun 19, 2026 (20:00 CEST)	8	0	1	8	28657
95	automated_watcher	changed	Jun 19, 2026 (22:00 CEST)	8	1	1	7	28657
96	automated_watcher	changed	Jun 20, 2026 (00:00 CEST)	8	1	1	7	28657
97	automated_watcher	changed	Jun 20, 2026 (05:00 CEST)	10	2	0	8	28657
98	automated_watcher	changed	Jun 20, 2026 (16:00 CEST)	11	1	0	10	28657
99	automated_watcher	changed	Jun 20, 2026 (21:00 CEST)	11	1	1	10	28657

#	Phase	Status	Source Time	Current	Added	Removed	Unchanged	Port
100	automated_watcher	changed	Jun 21, 2026 (00:00 CEST)	10	0	1	10	28657
101	automated_watcher	changed	Jun 21, 2026 (01:00 CEST)	11	1	0	10	28657
102	automated_watcher	changed	Jun 21, 2026 (09:00 CEST)	12	1	0	11	28657
103	automated_watcher	changed	Jun 21, 2026 (15:00 CEST)	11	0	1	11	28657
104	automated_watcher	changed	Jun 21, 2026 (22:00 CEST)	10	0	1	10	28657
105	automated_watcher	changed	Jun 21, 2026 (23:00 CEST)	10	1	1	9	28657
106	automated_watcher	changed	Jun 22, 2026 (09:00 CEST)	8	0	2	8	28657
107	automated_watcher	changed	Jun 22, 2026 (15:00 CEST)	6	0	2	6	28657
108	automated_watcher	changed	Jun 22, 2026 (19:00 CEST)	9	3	0	6	28657
109	automated_watcher	changed	Jun 22, 2026 (22:00 CEST)	10	2	1	8	28657
110	automated_watcher	changed	Jun 23, 2026 (00:00 CEST)	9	0	1	9	28657
111	automated_watcher	changed	Jun 23, 2026 (02:00 CEST)	10	1	0	9	28657
112	automated_watcher	changed	Jun 23, 2026 (06:00 CEST)	11	1	0	10	28657
113	automated_watcher	changed	Jun 23, 2026 (10:00 CEST)	12	1	0	11	28657
114	automated_watcher	changed	Jun 23, 2026 (16:00 CEST)	11	0	1	11	28657
115	automated_watcher	changed	Jun 23, 2026 (18:00 CEST)	10	0	1	10	28657
116	automated_watcher	changed	Jun 24, 2026 (01:00 CEST)	11	1	0	10	28657
117	automated_watcher	changed	Jun 24, 2026 (03:00 CEST)	12	1	0	11	28657
118	automated_watcher	changed	Jun 24, 2026 (07:00 CEST)	13	1	0	12	28657

#	Phase	Status	Source Time	Current	Added	Removed	Unchanged	Port
119	automated_watcher	changed	Jun 24, 2026 (14:00 CEST)	12	0	1	12	28657
120	automated_watcher	changed	Jun 24, 2026 (22:00 CEST)	11	0	1	11	28657
121	automated_watcher	changed	Jun 25, 2026 (00:00 CEST)	10	0	1	10	28657
122	automated_watcher	changed	Jun 25, 2026 (21:00 CEST)	7	0	3	7	28657
123	automated_watcher	changed	Jun 26, 2026 (02:00 CEST)	10	3	0	7	28657
124	automated_watcher	changed	Jun 26, 2026 (10:00 CEST)	11	1	0	10	28657
125	automated_watcher	changed	Jun 26, 2026 (21:00 CEST)	12	1	0	11	28657
126	automated_watcher	changed	Jun 27, 2026 (05:00 CEST)	10	0	2	10	28657
127	automated_watcher	changed	Jun 27, 2026 (08:00 CEST)	11	1	0	10	28657
128	automated_watcher	changed	Jun 27, 2026 (11:00 CEST)	12	1	0	11	28657
129	automated_watcher	changed	Jun 27, 2026 (16:00 CEST)	13	1	0	12	28657
130	automated_watcher	changed	Jun 28, 2026 (10:00 CEST)	12	0	1	12	28657
131	automated_watcher	changed	Jun 28, 2026 (12:00 CEST)	11	0	1	11	28657
132	automated_watcher	changed	Jun 28, 2026 (14:00 CEST)	12	1	0	11	28657
133	automated_watcher	changed	Jun 28, 2026 (19:00 CEST)	13	1	0	12	28657
134	automated_watcher	changed	Jun 30, 2026 (18:00 CEST)	14	1	0	13	28657
135	automated_watcher	changed	Jun 30, 2026 (22:00 CEST)	13	0	1	13	28657
136	automated_watcher	changed	Jul 1, 2026 (01:00 CEST)	12	0	1	12	28657
137	automated_watcher	changed	Jul 1, 2026 (12:00 CEST)	13	1	0	12	28657
138	automated_watcher	changed	Jul 1, 2026 (22:00 CEST)	14	1	0	13	28657

#	Phase	Status	Source Time	Current	Added	Removed	Unchanged	Port
139	automated_watcher	changed	Jul 2, 2026 (13:00 CEST)	15	1	0	14	28657
140	automated_watcher	changed	Jul 3, 2026 (19:00 CEST)	14	0	1	14	28657
141	automated_watcher	changed	Jul 4, 2026 (07:00 CEST)	13	0	1	13	28657
142	automated_watcher	changed	Jul 4, 2026 (18:00 CEST)	12	0	1	12	28657
143	automated_watcher	changed	Jul 8, 2026 (11:00 CEST)	13	1	0	12	28657
144	automated_watcher	changed	Jul 13, 2026 (08:00 CEST)	11	0	2	11	28657
145	automated_watcher	changed	Jul 14, 2026 (16:00 CEST)	10	0	1	10	28657
146	automated_watcher	changed	Jul 14, 2026 (17:00 CEST)	11	1	0	10	28657
147	automated_watcher	changed	Jul 15, 2026 (16:00 CEST)	12	1	0	11	28657
148	automated_watcher	changed	Jul 20, 2026 (15:00 CEST)	13	1	0	12	28657
149	automated_watcher	changed	Jul 23, 2026 (03:00 CEST)	14	1	0	13	28657
150	automated_watcher	changed	Jul 23, 2026 (13:00 CEST)	13	0	1	13	28657
151	automated_watcher	changed	2026-07-24 06:00 CEST	14	1	0	13	28657
152	automated_watcher	changed	2026-07-28 09:00 CEST	12	0	2	12	28657
153	automated_watcher	changed	2026-07-29 13:00 CEST	13	1	0	12	28657
154	automated_watcher	changed	2026-07-31 11:00 CEST	14	1	0	13	28657
155	automated_watcher	changed	2026-08-05 11:00 CEST	13	0	1	13	28657
156	automated_watcher	changed	2026-08-15 16:00 CEST	2	0	11	2	28657
157	automated_watcher	changed	2026-08-15 18:00 CEST	3	1	0	2	28657
158	automated_watcher	changed	2026-08-16 03:00 CEST	4	1	0	3	28657
159	automated_watcher	changed	2026-08-16 09:00 CEST	5	1	0	4	28657
160	automated_watcher	changed	2026-08-16 10:00 CEST	6	1	0	5	28657
161	automated_watcher	changed	2026-08-16 13:00 CEST	7	1	0	6	28657
162	automated_watcher	changed	2026-08-17 03:00 CEST	8	1	0	7	28657
163	automated_watcher	changed	2026-08-18 03:00 CEST	9	1	0	8	28657

#	Phase	Status	Source Time	Current	Added	Removed	Unchanged	Port
164	automated_watcher	changed	2026-08-18 05:00 CEST	10	1	0	9	28657
165	automated_watcher	changed	2026-08-19 16:00 CEST	9	0	1	9	28657
166	automated_watcher	changed	2026-08-19 21:00 CEST	10	1	0	9	28657
167	automated_watcher	changed	2026-08-20 18:00 CEST	9	0	1	9	28657
168	automated_watcher	changed	2026-08-21 03:00 CEST	8	0	1	8	28657
169	automated_watcher	changed	2026-08-21 06:00 CEST	9	1	0	8	28657
170	automated_watcher	changed	2026-08-22 06:00 CEST	8	0	1	8	28657
171	automated_watcher	changed	2026-08-22 11:00 CEST	9	1	0	8	28657
172	automated_watcher	changed	2026-08-23 06:00 CEST	8	0	1	8	28657
173	automated_watcher	changed	2026-08-23 12:00 CEST	6	0	2	6	28657
174	automated_watcher	changed	2026-08-23 12:00 CEST	8	2	0	6	28657
175	automated_watcher	changed	2026-08-25 03:00 CEST	9	1	0	8	28657
176	automated_watcher	changed	2026-08-25 05:00 CEST	10	1	0	9	28657
177	automated_watcher	changed	2026-08-26 21:00 CEST	11	1	0	10	28657
178	automated_watcher	changed	2026-08-28 23:00 CEST	10	0	1	10	28657
179	automated_watcher	changed	2026-08-29 12:00 CEST	9	0	1	9	28657
180	automated_watcher	changed	2026-08-30 09:00 CEST	10	1	0	9	28657
181	automated_watcher	changed	2026-08-30 15:00 CEST	9	0	1	9	28657
182	automated_watcher	changed	2026-08-30 19:00 CEST	10	1	0	9	28657
183	automated_watcher	changed	2026-09-03 13:00 CEST	6	1	5	5	28657
184	automated_watcher	changed	2026-09-03 17:00 CEST	7	2	1	5	28657
185	automated_watcher	changed	2026-09-04 20:00 CEST	6	0	1	6	28657
186	automated_watcher	changed	2026-09-06 18:00 CEST	7	1	0	6	28657
187	automated_watcher	changed	2026-09-06 20:00 CEST	6	0	1	6	28657
188	automated_watcher	changed	2026-09-08 02:00 CEST	6	1	1	5	28657
189	automated_watcher	changed	2026-09-08 07:00 CEST	7	1	0	6	28657
190	automated_watcher	changed	2026-09-09 17:00 CEST	6	0	1	6	28657
191	automated_watcher	changed	2026-09-09 20:00 CEST	7	1	0	6	28657

#	Phase	Status	Source Time	Current	Added	Removed	Unchanged	Port
192	automated_watcher	changed	2026-09-11 08:00 CEST	5	0	2	5	28657
193	automated_watcher	changed	2026-09-11 13:00 CEST	6	1	0	5	28657
194	automated_watcher	changed	2026-09-12 08:00 CEST	4	0	2	4	28657
195	automated_watcher	changed	2026-09-12 13:00 CEST	6	2	0	4	28657

7. Report-Use Notes

- Use the 7D report for short-range rotation and recent watcher movement.
- Use the 30D report for broader trend context once the 15-minute watcher has accumulated enough observations.
- Use the All Time report for full testnet observation history preserved by this watcher.
- For publication-quality interpretation, compare this Markdown output with dashboard charts, raw JSON outputs, and selected snapshots.